

Операционные системы

Командная оболочка Midnight Commander

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Цели и задачи работы

Ознакомление с файловой системой Linux, её структурой, именами и содержанием каталогов. Приобретение практических навыков по применению команд для работы с файлами и каталогами, по управлению процессами, по проверке использования диска и обслуживанию файловой системы.

- 1 Изучить возможности Midnight Commander
- 2 Изучить редактор Midnight Commander

Процесс выполнения лабораторной работы

Работа с Midnight Commander

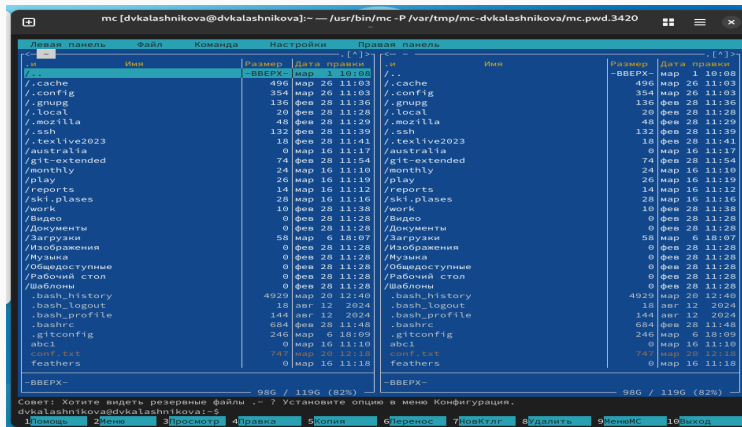


Рис. 1: Запуск mc

Работа с Midnight Commander

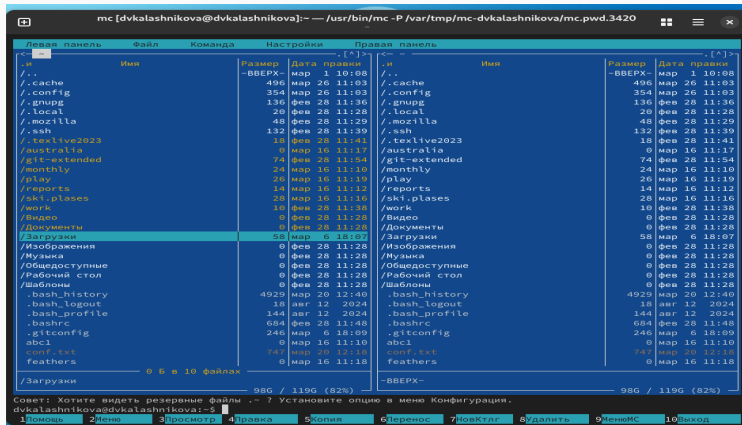


Рис. 2: Выделение

Работа с Midnight Commander

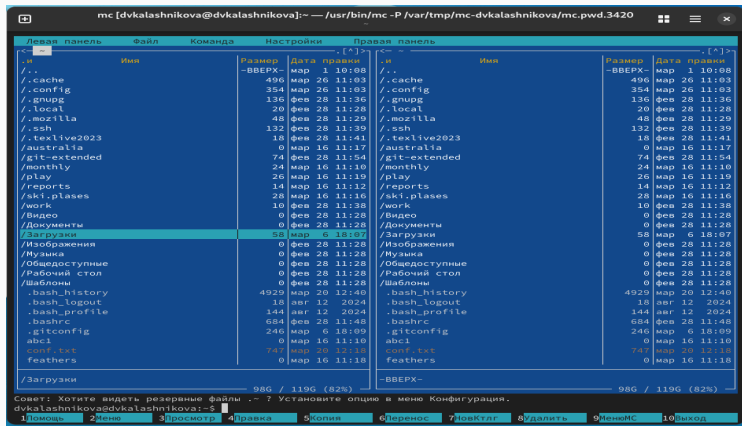


Рис. 3: Отмена

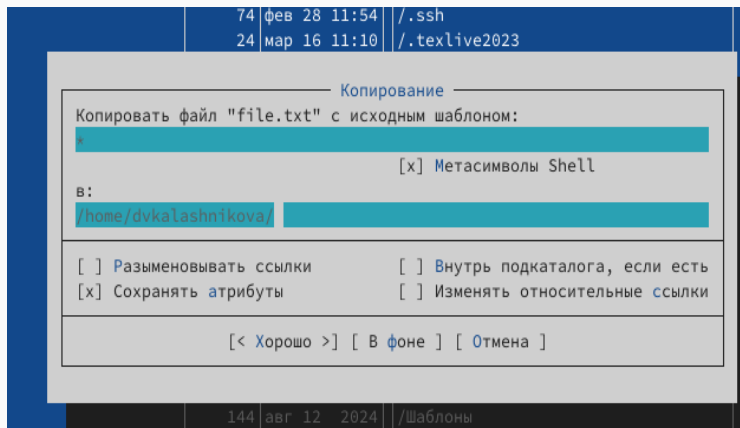


Рис. 4: Копирование

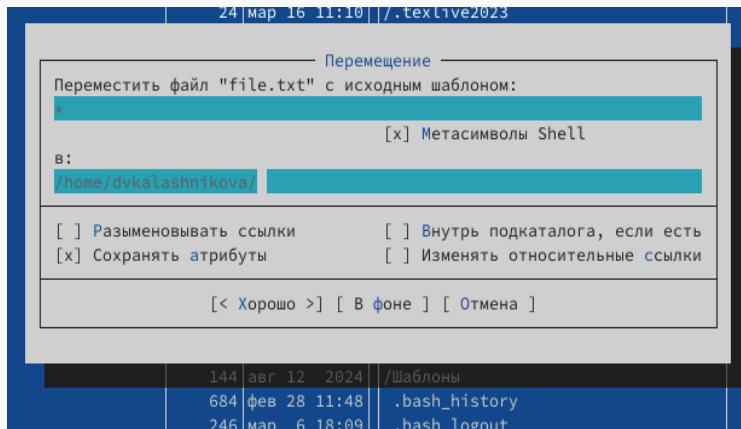


Рис. 5: Перемещение

Работа с Midnight Commander

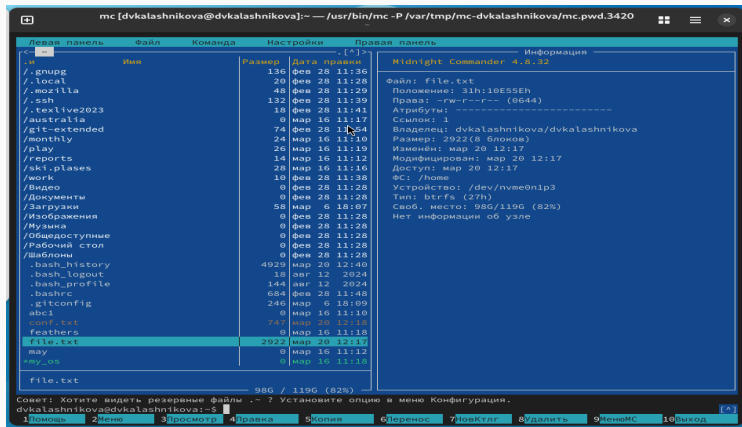


Рис. 6: Информация

Работа с Midnight Commander

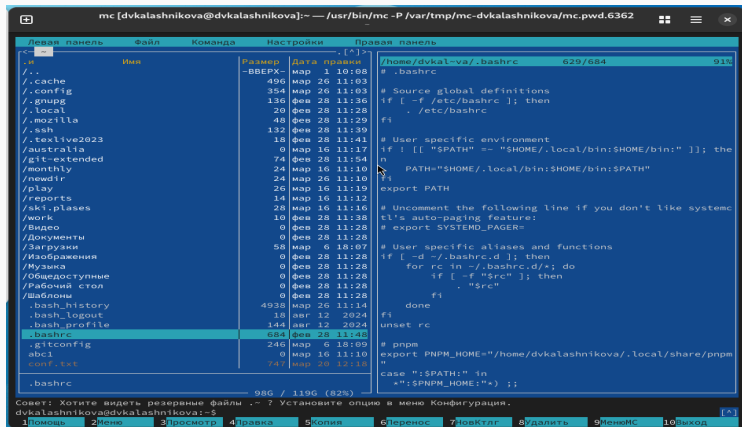


Рис. 7: Быстрый просмотр

Работа с Midnight Commander

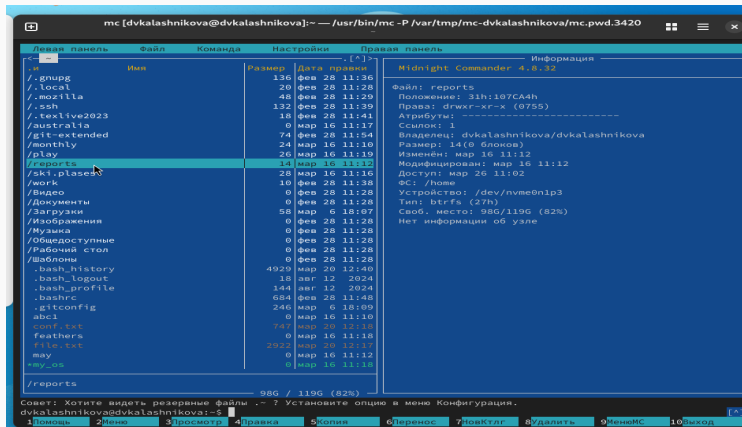


Рис. 8: Информация

Работа с Midnight Commander

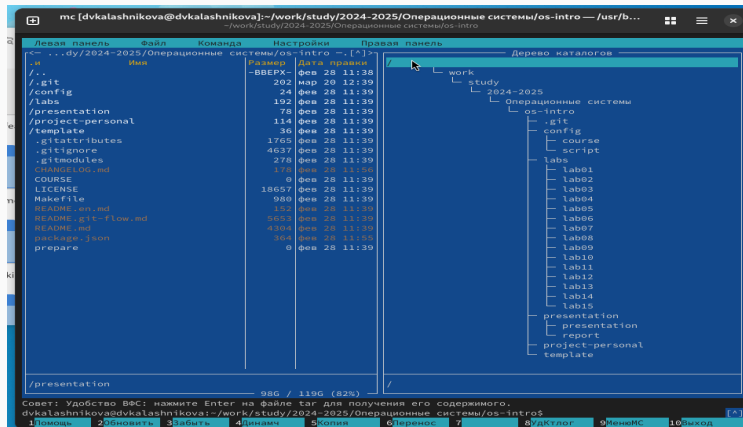
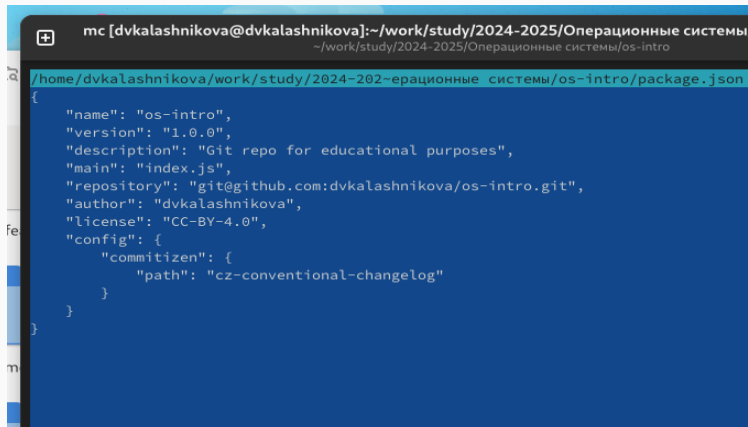


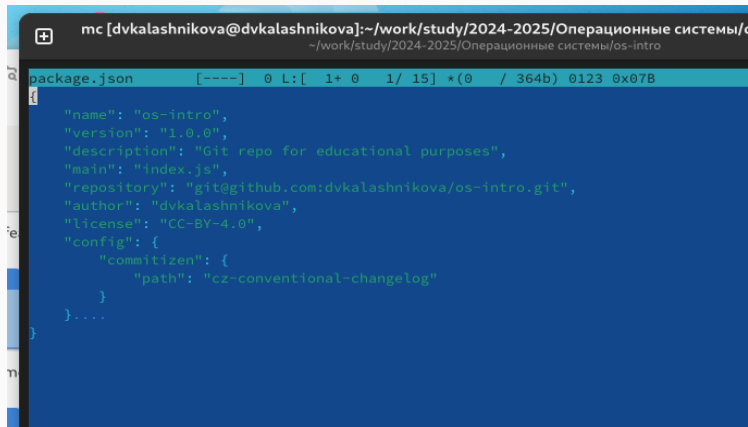
Рис. 9: Дерево каталогов



The screenshot shows the Midnight Commander (mc) interface. The title bar at the top reads "mc [dvkashnikova@dvkashnikova]:~/work/study/2024-2025/Операционные системы". Below the title bar, the current directory is shown as "~/work/study/2024-2025/Операционные системы/os-intro". The main window displays the contents of the file "package.json" located at "/home/dvkalashnikova/work/study/2024-2025-ерационные системы/os-intro/package.json". The file content is a JSON object with the following fields: "name", "version", "description", "main", "repository", "author", "license", and "config". The "config" field contains a "commitizen" object with a "path" field.

```
{
  "name": "os-intro",
  "version": "1.0.0",
  "description": "Git repo for educational purposes",
  "main": "index.js",
  "repository": "git@github.com:dvkashnikova/os-intro.git",
  "author": "dvkashnikova",
  "license": "CC-BY-4.0",
  "config": {
    "commitizen": {
      "path": "cz-conventional-changelog"
    }
  }
}
```

Рис. 10: Просмотр содержимого текстового файла



The screenshot shows the Midnight Commander (mc) interface. The title bar at the top reads "mc [dvkalashnikova@dvkalashnikova]:~/work/study/2024-2025/Операционные системы/с" and the path below it is "~/work/study/2024-2025/Операционные системы/os-intro". The main window displays the content of the "package.json" file. The file content is a JSON object with the following fields: "name" (os-intro), "version" (1.0.0), "description" (Git repo for educational purposes), "main" (index.js), "repository" (git@github.com:dvkalashnikova/os-intro.git), "author" (dvkalashnikova), "license" (CC-BY-4.0), and "config" (an object with "commitizen" which has a "path" of "cz-conventional-changelog"). The JSON is partially visible, showing the opening and closing braces and ellipses. The interface has a dark blue background with light blue text.

```
package.json  [----]  0 L:[ 1+ 0  1/ 15] *(0  / 364b) 0123 0x07B
{
  "name": "os-intro",
  "version": "1.0.0",
  "description": "Git repo for educational purposes",
  "main": "index.js",
  "repository": "git@github.com:dvkalashnikova/os-intro.git",
  "author": "dvkalashnikova",
  "license": "CC-BY-4.0",
  "config": {
    "commitizen": {
      "path": "cz-conventional-changelog"
    }
  }...
}
```

Рис. 11: Отредактируем содержимое текстового файла без сохранения результатов

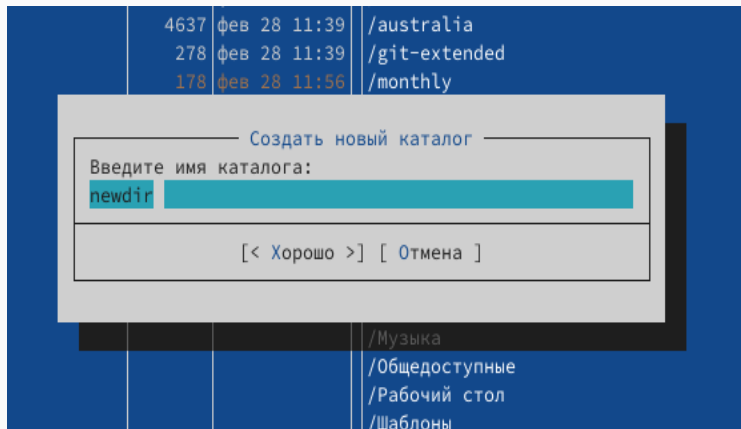


Рис. 12: Создание каталога

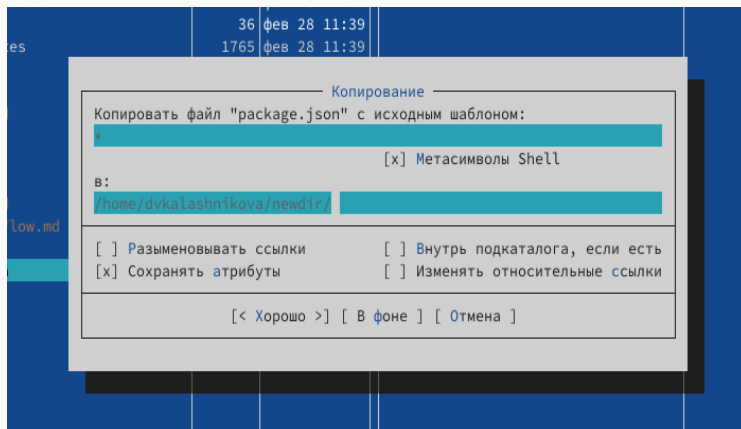


Рис. 13: Копирование в файлов в созданный каталог

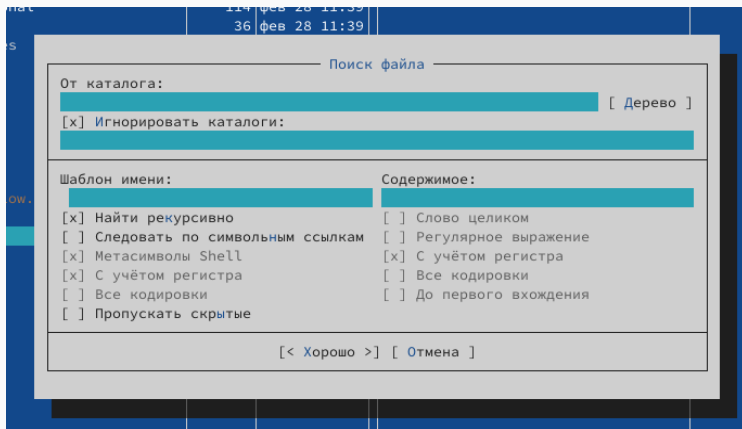


Рис. 14: Поиск файлов

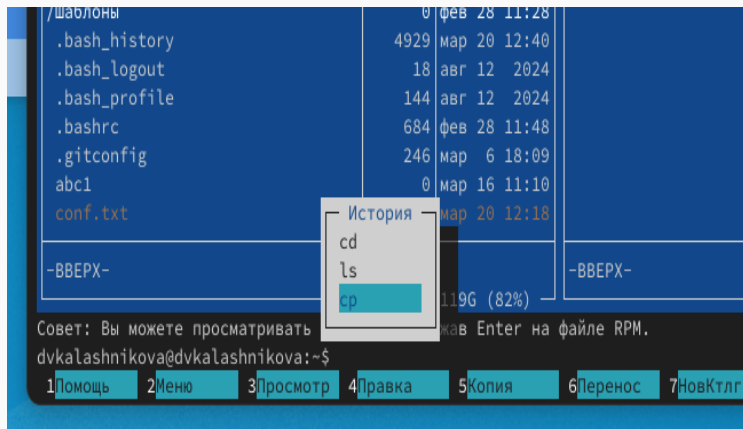


Рис. 15: История команд

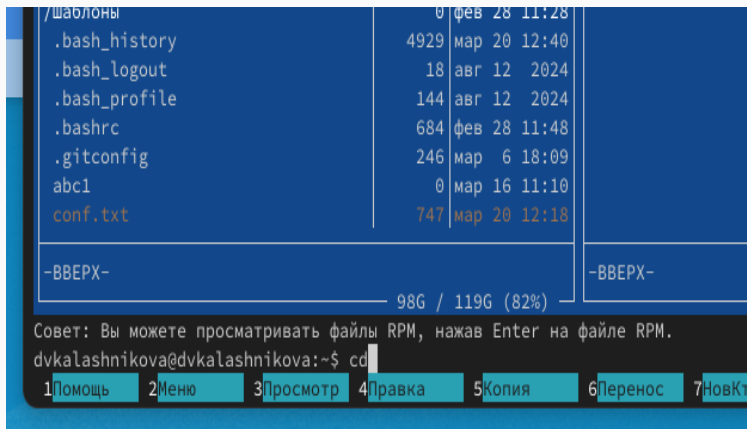
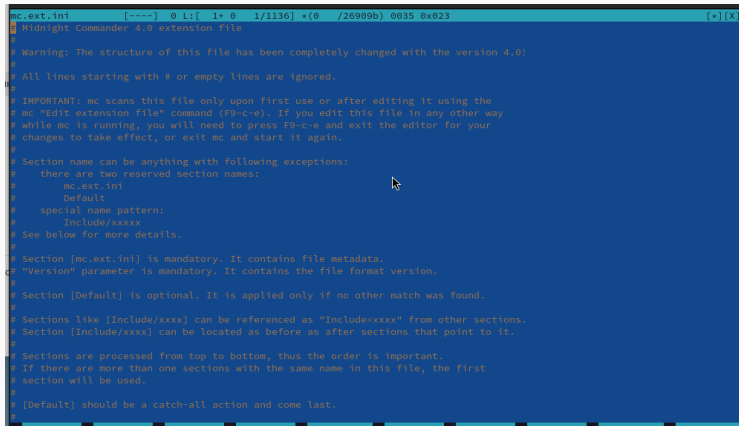


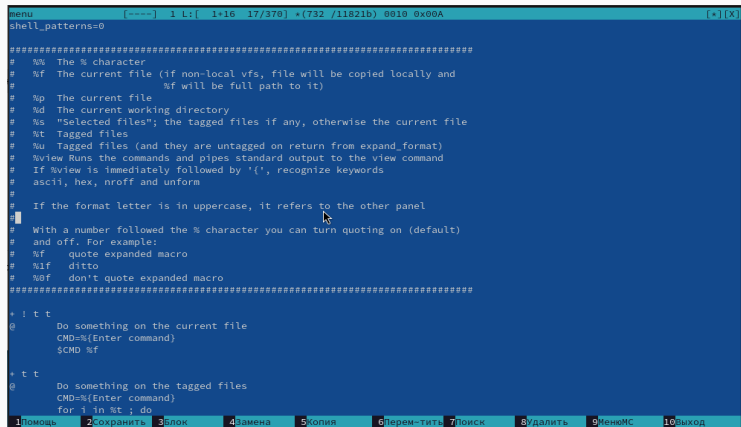
Рис. 16: Переход в домашний каталог



```
mc.ext.ini [----] 0 L:[ 1+ 0 1/1136]*(0 /26909b) 0035 0x023 [*][X]
# Midnight Commander 4.0 extension file

# Warning: The structure of this file has been completely changed with the version 4.0!
#
# All lines starting with # or empty lines are ignored.
#
# IMPORTANT: mc scans this file only upon first use or after editing it using the
# mc "Edit extension file" command (F9-c-e). If you edit this file in any other way
# while mc is running, you will need to press F9-c-e and exit the editor for your
# changes to take effect, or exit mc and start it again.
#
# Section name can be anything with following exceptions:
#   there are two reserved section names:
#       mc.ext.ini
#       Default
#   special name pattern:
#       Include/xxxxx
# See below for more details.
#
# Section [mc.ext.ini] is mandatory. It contains file metadata.
# "Version" parameter is mandatory. It contains the file format version.
#
# Section [Default] is optional. It is applied only if no other match was found.
#
# Sections like [Include/xxxx] can be referenced as "Include=xxxx" from other sections.
# Section [Include/xxxx] can be located as before as after sections that point to it.
#
# Sections are processed from top to bottom, thus the order is important.
# If there are more than one sections with the same name in this file, the first
# section will be used.
#
# [Default] should be a catch-all action and come last.
```

Рис. 17: Просмотр файла расширений

A screenshot of the Midnight Commander file manager's menu system. The window has a title bar with a menu icon, a status bar showing "1 L: [1*16 17/370] *(732 /11821b) 0010 0x00A", and a close button. The main area displays the "menu" and "shell_patterns=0" text, followed by a list of macros. The macros include explanations for %%, %f, %p, %d, %s, %t, %u, %view, and %! commands. At the bottom, a menu bar contains 10 numbered items: 1.Помощь, 2.Сохранить, 3.Блок, 4.Замена, 5.Копия, 6.Перем-тить, 7.Поиск, 8.Удалить, 9.МенюМС, 10.Выход.

```
menu
shell_patterns=0

#####
# %% The % character
# %f The current file (if non-local vfs, file will be copied locally and
#    %f will be full path to it)
# %p The current file
# %d The current working directory
# %s "Selected files"; the tagged files if any, otherwise the current file
# %t Tagged files
# %u Tagged files (and they are untagged on return from expand_format)
# %view Runs the commands and pipes standard output to the view command
# If %view is immediately followed by '{', recognize keywords
#    ascii, hex, nroff and unform
#
# If the format letter is in uppercase, it refers to the other panel
#
# With a number followed the % character you can turn quoting on (default)
# and off. For example:
# %f quote expanded macro
# %!f ditto
# %!0f don't quote expanded macro
#####

+ ! t t
@    Do something on the current file
    CMD=%{Enter command}
    $CMD %f

+ t t
@    Do something on the tagged files
    CMD=%{Enter command}
    for i in %t ; do
```

Рис. 18: Просмотр файла меню

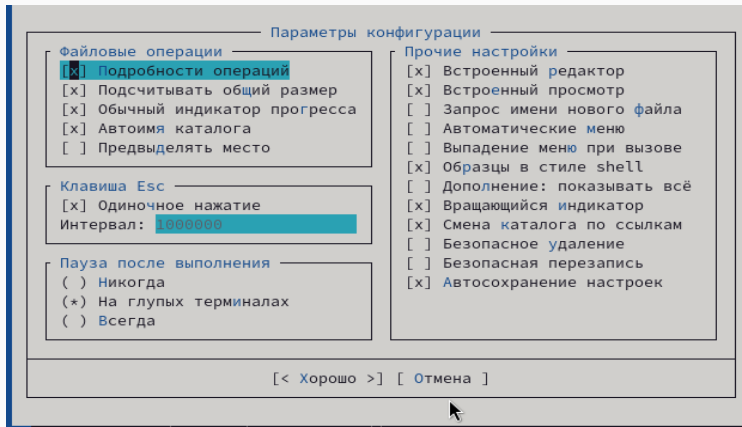


Рис. 19: Конфигурация

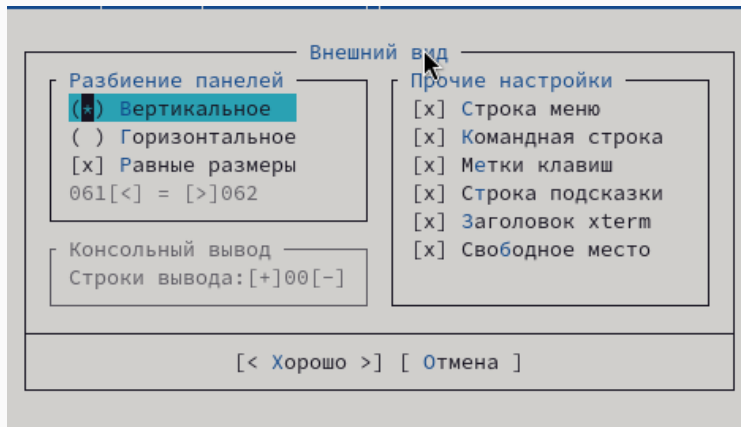


Рис. 20: Внешний вид

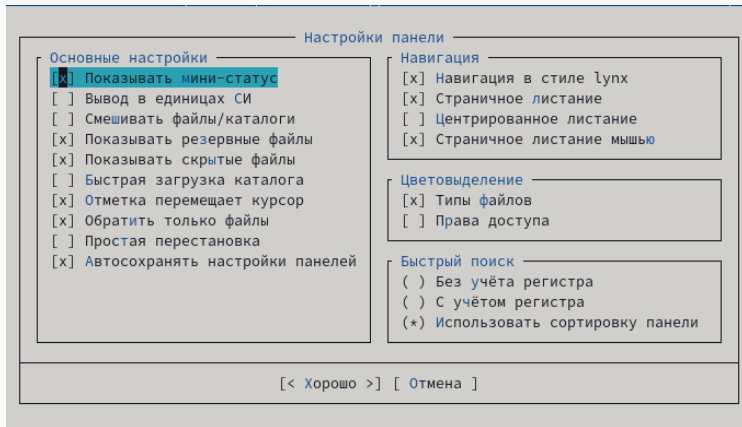


Рис. 21: Настройки панелей

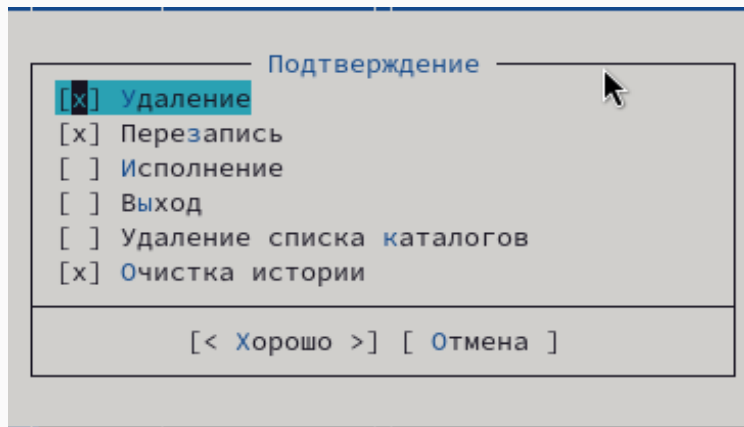


Рис. 22: Подтверждение

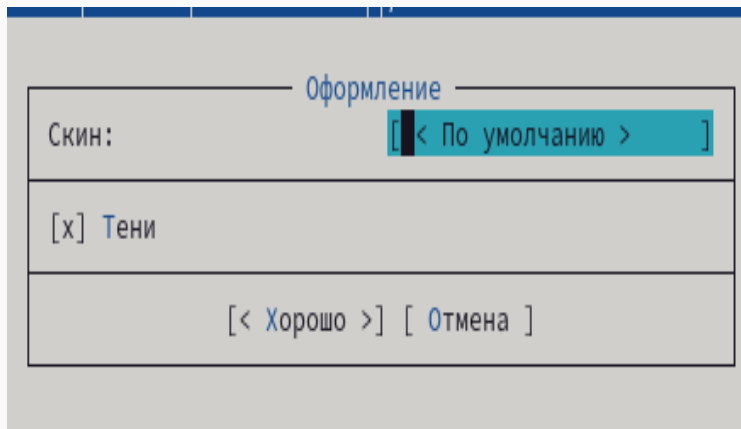


Рис. 23: Оформление

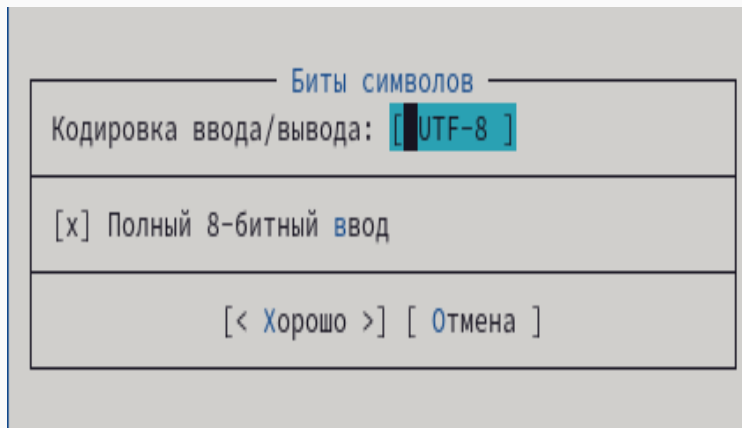


Рис. 24: Кодировка символов

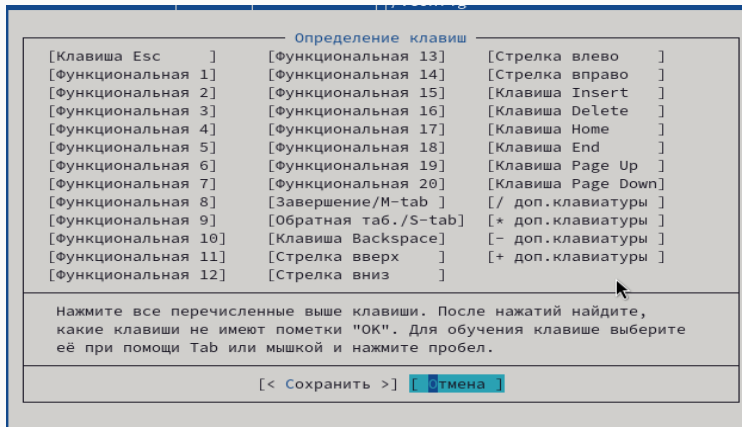
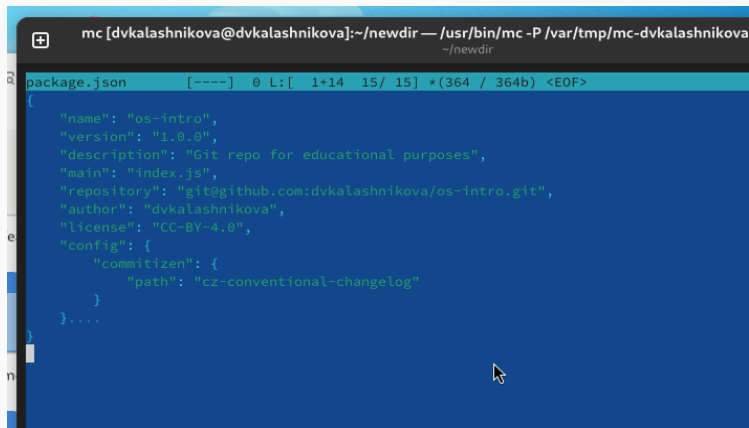


Рис. 25: Распознавание клавиш

Работа с редактором Midnight Commander

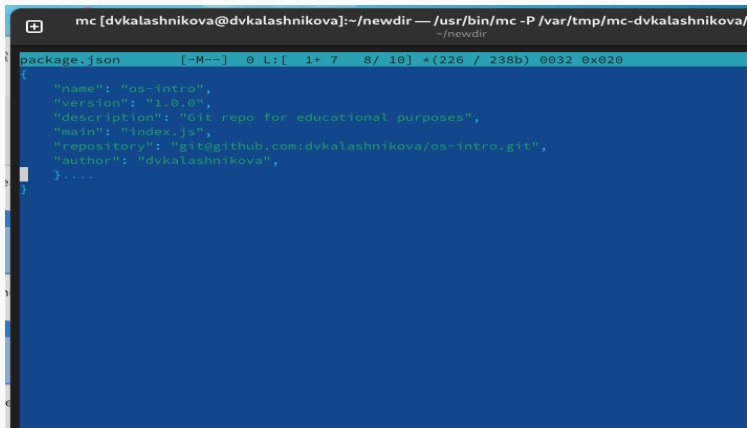


The screenshot shows the Midnight Commander (MC) file editor interface. The title bar at the top indicates the user is 'dvkalashnikova' and the current directory is '~/newdir'. The status bar shows the file 'package.json' with a size of 364 bytes. The main editing area displays the content of the package.json file, which is a JSON object with fields for name, version, description, main, repository, author, license, and config. The config field contains a commitizen configuration. The text is displayed in a monospaced font on a dark blue background.

```
mc [dvkalashnikova@dvkalashnikova]:~/newdir — /usr/bin/mc -P /var/tmp/mc-dvkalashnikova
~/newdir

package.json [----] 0 L: [ 1+14 15/ 15] *(364 / 364b) <EOF>
{
  "name": "os-intro",
  "version": "1.0.0",
  "description": "Git repo for educational purposes",
  "main": "index.js",
  "repository": "git@github.com:dvkalashnikova/os-intro.git",
  "author": "dvkalashnikova",
  "license": "CC-BY-4.0",
  "config": {
    "commitizen": {
      "path": "cz-conventional-changelog"
    }
  }
}....
```

Рис. 26: Файл с текстом



The screenshot shows the Midnight Commander (MC) interface. The title bar at the top reads: `mc [dvkalashnikova@dvkalashnikova]:~/newdir — /usr/bin/mc -P /var/tmp/mc-dvkalashnikova/`. Below the title bar, the file `package.json` is open, with a status line showing `[-M--] 0 L:[ 1+ 7 8/ 10] *(226 / 238b) 0032 0x020`. The main editing area displays the following JSON content:

```
{  
  "name": "os-intro",  
  "version": "1.0.0",  
  "description": "Git repo for educational purposes",  
  "main": "index.js",  
  "repository": "git@github.com:dvkalashnikova/os-intro.git",  
  "author": "dvkalashnikova",  
  }....  
}
```

Рис. 27: Файл с текстом

[illegible]

Рис. 28: Копирование фрагмента

Работа с редактором Midnight Commander

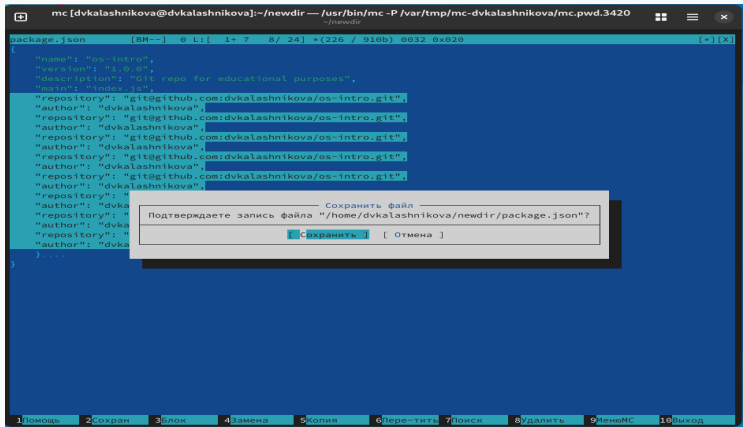


Рис. 29: Сохранение

Работа с редактором Midnight Commander

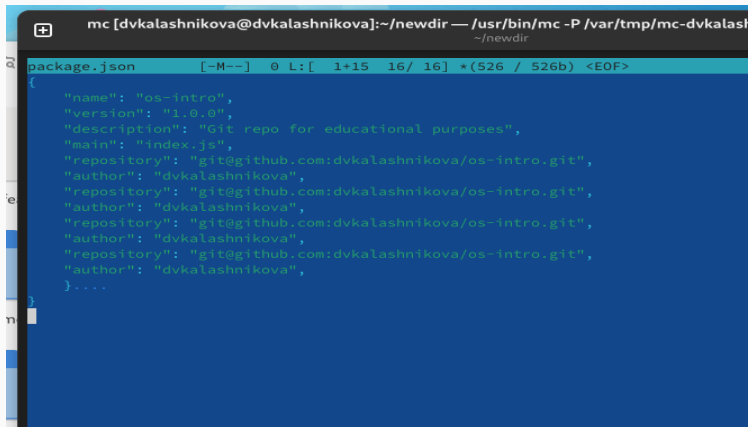
```
mc [dvkalashnikova@dvkalashnikova]:~/newdir — /usr/bin/mc -P /var/tmp/mc-dvkalashnikova/mc.pwd.3420 ~/newdir
```

```
package.json    [BM--]   0 L:[ 1+ 7   8/ 16] *(226 / 526b) 0032 0x020
```

```
{  
  "name": "os-intro",  
  "version": "1.0.0",  
  "description": "Git repo for educational purposes",  
  "main": "index.js",  
  "repository": "git@github.com:dvkalashnikova/os-intro.git",  
  "author": "dvkalashnikova",  
  "repository": "git@github.com:dvkalashnikova/os-intro.git",  
  "author": "dvkalashnikova",  
  "repository": "git@github.com:dvkalashnikova/os-intro.git",  
  "author": "dvkalashnikova",  
  "repository": "git@github.com:dvkalashnikova/os-intro.git",  
  "author": "dvkalashnikova",  
  "repository": "git@github.com:dvkalashnikova/os-intro.git",  
  "author": "dvkalashnikova",  
}, ...  
}
```

Рис. 30: Отмена

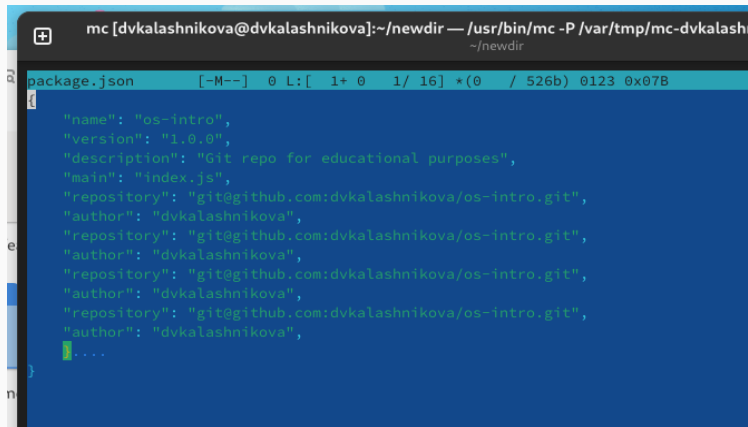
Работа с редактором Midnight Commander



The screenshot shows the Midnight Commander (MC) interface. The title bar at the top reads "mc [dvkalashnikova@dvkalashnikova]:~/newdir — /usr/bin/mc -P /var/tmp/mc-dvkalashnikova ~/newdir". The current file being edited is "package.json", and the status bar indicates it is at line 16, column 16, with a cursor at the end of the file. The file content is a JSON object for an npm package named "os-intro". The visible part of the file is as follows:

```
{
  "name": "os-intro",
  "version": "1.0.0",
  "description": "Git repo for educational purposes",
  "main": "index.js",
  "repository": "git@github.com:dvkalashnikova/os-intro.git",
  "author": "dvkalashnikova",
  "repository": "git@github.com:dvkalashnikova/os-intro.git",
  "author": "dvkalashnikova",
  "repository": "git@github.com:dvkalashnikova/os-intro.git",
  "author": "dvkalashnikova",
  "repository": "git@github.com:dvkalashnikova/os-intro.git",
  "author": "dvkalashnikova",
  }....
```

Рис. 31: Переход в конец файла

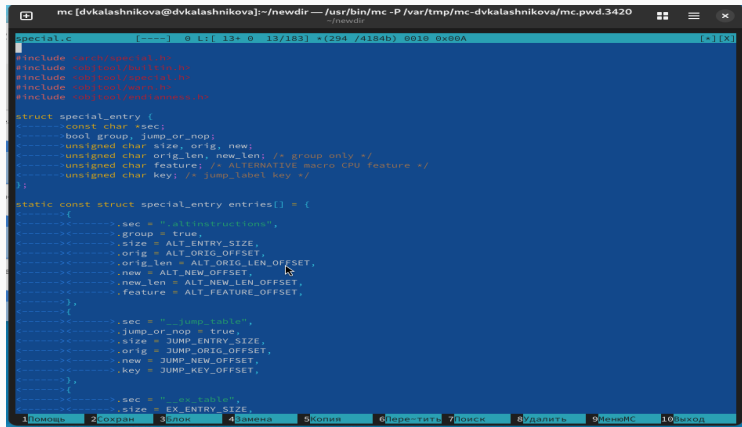


The screenshot shows the Midnight Commander (MC) interface. The title bar at the top reads "mc [dvkalashnikova@dvkalashnikova]:~/newdir — /usr/bin/mc -P /var/tmp/mc-dvkalashnikova ~/newdir". The main window displays the "package.json" file. The file's metadata bar shows "[-M--] 0 L: [1+ 0 1/ 16] *(0 / 526b) 0123 0x07B". The file content is a JSON object with the following fields: "name": "os-intro", "version": "1.0.0", "description": "Git repo for educational purposes", "main": "index.js", and several "repository" and "author" entries. The cursor is positioned at the beginning of the first line of the JSON object, which is "{".

```
{
  "name": "os-intro",
  "version": "1.0.0",
  "description": "Git repo for educational purposes",
  "main": "index.js",
  "repository": "git@github.com:dvkalashnikova/os-intro.git",
  "author": "dvkalashnikova",
  "repository": "git@github.com:dvkalashnikova/os-intro.git",
  "author": "dvkalashnikova",
  "repository": "git@github.com:dvkalashnikova/os-intro.git",
  "author": "dvkalashnikova",
  "repository": "git@github.com:dvkalashnikova/os-intro.git",
  "author": "dvkalashnikova",
  ...
}
```

Рис. 32: Переход в начало файла

Работа с редактором Midnight Commander



```
mc [dvkashnikova@dvkashnikova]:~/newdir — /usr/bin/mc -P /var/tmp/mc-dvkashnikova/mc.pwd.3420
~newdir
special.c [---] 0 L: 13+ 0 13/183 *(294 /4184b) 0010 0x00A [*] [X]

#include <stdio.h>
#include <stdlib.h>
#include <string.h>
#include <unistd.h>
#include <sys/types.h>

struct special_entry {
    const char *sec;
    bool group, jump_or_nop;
    unsigned char size, orig, new;
    unsigned char orig_len, new_len; /* group only */
    unsigned char feature; /* ALTERNATIVE macro CPU feature */
    unsigned char key; /* jump_label key */
};

static const struct special_entry entries[] = {
    {
        .sec = "alt_instructions",
        .group = true,
        .size = ALT_ENTRY_SIZE,
        .orig = ALT_ORIG_OFFSET,
        .orig_len = ALT_ORIG_LEN_OFFSET,
        .new = ALT_NEW_OFFSET,
        .new_len = ALT_NEW_LEN_OFFSET,
        .feature = ALT_FEATURE_OFFSET,
    },
    {
        .sec = "__jump_table",
        .jump_or_nop = true,
        .size = JUMP_ENTRY_SIZE,
        .orig = JUMP_ORIG_OFFSET,
        .new = JUMP_NEW_OFFSET,
        .key = JUMP_KEY_OFFSET,
    },
    {
        .sec = "__ex_table",
        .size = EX_ENTRY_SIZE,
    }
};
```

1Помощь 2Сохран 3Блок 4Замена 5Копия 6Пере-тить 7Поиск 8Удалить 9МенюМС 10Выход

Рис. 33: Файл с программой

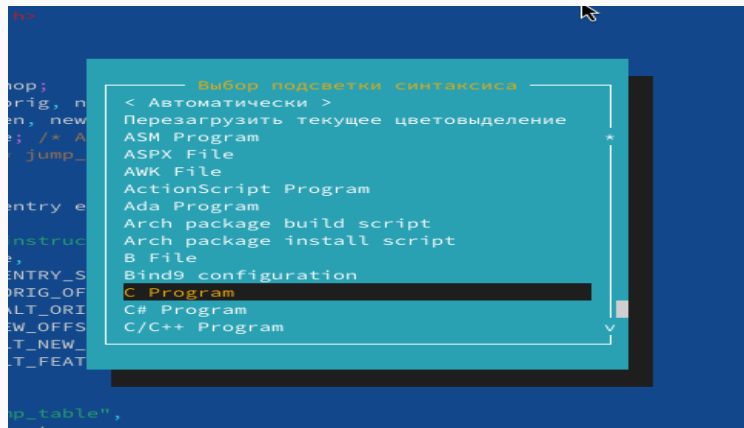


Рис. 34: Цветовыделение синтаксиса

Выводы по проделанной работе

В данной работе мы ознакомились с инструментами поиска файлов и фильтрации текстовых данных. А также приобрели практические навыки по управлению процессами.